

ESPERANCE, Australia

WMO: 946380

Lat: 33.8300S

Lon: 121.8925E

Elev: 27

StdP: 101.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	7.0	-1.2	3.4	19.2	0.5	3.9	17.1	12.8	14.7	11.5	15.1	4.1	10	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.5	33.7	18.0	30.4	17.7	27.7	17.5	20.7	26.8	19.9	25.2	19.3	24.1	6.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	21.7	18.4	13.3	21.4	17.7	12.7	21.1	59.9	27.0	57.1	25.2	55.1	24.1	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	10.1	9.2	DB	4.3	41.1	0.8	1.9	3.7	42.4	3.3	43.5	2.8	44.6	2.2	46.0	(4)
(5)			WB	3.4	22.5	0.8	1.3	2.8	23.4	2.4	24.2	1.9	25.0	1.3	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	20.3	20.5	19.6	17.9	15.6	13.5	12.7	13.2	14.2	15.6	17.4	18.8	(6)	
	DBStd	3.80	2.95	2.78	2.93	2.83	2.40	1.83	1.82	2.25	2.49	2.94	3.13	3.03	(7)	
	HDD10.0	3	0	0	0	0	0	1	1	1	1	0	0	0	(8)	
	HDD18.3	949	7	5	17	41	92	146	176	162	127	97	54	26	(9)	
	CDD10.0	2403	319	294	297	238	173	105	84	98	127	174	222	271	(10)	
	CDD18.3	308	68	66	55	29	7	1	0	1	3	13	27	39	(11)	
	CDH23.3	2069	442	361	329	177	43	2	1	17	42	131	231	294	(12)	
(13)	CDH26.7	820	191	143	129	56	7	0	0	3	11	50	99	130	(13)	
(14)	Wind	WSAvg	5.1	5.6	5.5	5.0	4.5	4.5	4.9	5.1	5.0	5.2	5.2	5.4	5.5	(14)
(15) Precipitation	PrecAvg	611	17	26	30	45	72	77	97	85	62	47	34	19	(15)	
	PrecMax	868	90	146	148	155	186	193	192	149	120	117	88	62	(16)	
	PrecMin	446	0	0	1	3	15	29	23	27	16	6	1	1	(17)	
	PrecStd	107	17	28	32	31	37	33	35	30	29	25	21	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	36.1	35.2	32.0	27.5	22.5	22.1	25.7	27.7	32.3	34.7	36.6	(19)	
		MCWB	19.6	19.4	18.6	16.7	15.5	13.3	13.1	14.1	14.6	15.4	17.0	18.1	(20)	
	2%	DB	31.6	31.0	30.2	27.8	24.1	20.4	19.7	21.8	23.7	27.2	29.3	30.0	(21)	
		MCWB	19.1	19.2	17.9	16.4	14.9	12.8	12.4	13.2	14.0	15.2	16.4	17.8	(22)	
	5%	DB	27.5	27.4	26.8	24.9	21.9	18.9	18.1	19.5	21.0	23.1	25.2	26.0	(23)	
		MCWB	19.0	19.0	17.8	16.5	14.6	12.8	12.0	12.7	13.6	15.0	16.4	17.8	(24)	
	10%	DB	24.9	24.8	24.1	22.5	20.1	17.6	16.7	17.6	18.9	20.6	22.5	23.3	(25)	
		MCWB	18.4	18.7	17.8	16.3	14.3	12.3	11.7	12.2	13.2	14.5	16.2	17.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.3	20.7	19.3	18.1	15.8	14.7	15.5	16.5	17.8	19.5	20.3	(27)	
		MCDB	29.2	28.1	26.1	23.8	22.1	19.2	18.1	20.6	22.6	23.8	26.3	27.6	(28)	
	2%	WB	20.6	20.9	19.7	18.5	16.9	14.8	13.8	14.6	15.3	16.6	18.2	19.2	(29)	
		MCDB	27.6	26.2	24.4	22.5	20.1	17.7	17.1	18.8	19.9	22.4	24.2	25.6	(30)	
	5%	WB	19.7	20.0	19.2	17.8	16.0	13.9	13.1	13.7	14.6	15.7	17.4	18.3	(31)	
		MCDB	25.4	24.8	23.6	21.6	19.6	16.9	16.4	17.7	18.8	21.0	22.6	23.9	(32)	
	10%	WB	18.9	19.3	18.6	17.2	15.2	13.1	12.3	12.9	13.8	15.0	16.6	17.5	(33)	
		MCDB	23.8	23.5	22.8	21.0	18.6	16.3	15.6	16.6	17.9	19.7	21.4	22.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.5	8.8	9.2	8.7	8.2	8.1	8.8	9.1	9.2	9.1	8.5	(35)	
		MCDBR	15.2	14.6	14.4	13.4	11.8	10.6	10.7	12.5	13.5	15.2	15.7	14.9	(36)	
	5% WB	MCWBR	5.6	5.4	5.1	5.4	5.3	5.5	5.5	6.0	6.0	6.4	6.1	6.0	(37)	
		MCWBR	12.0	10.8	10.2	9.5	9.1	8.4	8.7	10.3	11.0	11.9	12.4	12.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.360	0.354	0.336	0.326	0.308	0.299	0.299	0.309	0.323	0.339	0.344	0.342	(40)	
		taud	2.527	2.554	2.594	2.615	2.635	2.630	2.626	2.589	2.544	2.516	2.526	2.554	(41)	
	Ebn at Noon		977	962	945	900	866	849	868	905	943	967	987	998	(42)	
		Edh at Noon	111	104	94	82	71	67	70	82	96	107	110	109	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.30	6.37	5.25	3.95	2.91	2.48	2.69	3.60	4.83	6.01	6.89	7.54	(44)	
	RadStd		0.40	0.44	0.35	0.19	0.17	0.09	0.14	0.17	0.19	0.37	0.45	0.46	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.22	N/A	-0.68	N/A	N/A	N/A	N/A	-60	+93	+44	(46)
(47)	Regional (6 neighbors)	+0.23	N/A	N/A	N/A	N/A	N/A	N/A	-60	+93	+44	(47)

Nomenclature: See separate page